

JEE MAIN REVISION MASTER: CHEMISTRY IN EVERYDAY LIFE OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 77 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Chemistry In Everyday Life — Standard Calculations & Problem Archetypes	35 PYQs	Quantitative Analysis
Very High	Chemistry In Everyday Life — Applied Cases, Reactions & Traps	30 PYQs	Applied Topics
Very High	Chemistry In Everyday Life — Core Principles & Governing Laws	15 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Chemistry In Everyday Life — Standard Calculations & Problem Archetypes	$C_{14}H_{25}O_4N_3S$ - \text{Alitame}
Chemistry In Everyday Life — Applied Cases, Reactions & Traps	$C_{17}H_{35}COO^-Na^+ + Na_2CO_3$, $CH_3(CH_2)_{16}COO^-Na^+$, $CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$
Chemistry In Everyday Life — Core Principles & Governing Laws	$(C_{11}H_{23}CH_2OSO_3^-Na^+)$, $(C_{17}H_{35}COO^-Na^+)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Chemistry In Everyday Life — Standard Calculations & Problem Archetypes (Very High Yield — 35 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Morning Shift] Q#1 (MCQ): "Match List I with List II LIST I (Compound) LIST II (Uses) A. Iodoform I. Fire extinguisher B. Carbon tetrachloride II. Insecticide C. CFC II..." [Ans: B]
- [2023 Morning Shift] Q#1 (MCQ): "Match List I with List II: LIST I LIST II A. Saccharin I. High potency sweetener B. Aspartame II. First artificial sweetening agent C. Alita..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Match List I with List II List I List II A. Tranquilizers I. Anti blood clotting B. Aspirin II. Salvarsan C. Antibiotic III. antidepressant ..." [Ans: N/A]

3.2 Chemistry In Everyday Life — Applied Cases, Reactions & Traps (Very High Yield — 30 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Morning Shift] Q#1 (MCQ): "Match List I with List II LIST I (Compound) LIST II (Uses) A. Iodoform I. Fire extinguisher B. Carbon tetrachloride II. Insecticide C. CFC II..." [Ans: B]
- [2024 Evening Shift] Q#2 (MCQ): "Match List - I with List - II. List I (Compound) List II (Use) (A) Carbon tetrachloride (I) Paint remover (B) Methylene chloride (II) Refrig..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Amongst the following compounds, which one is an antacid?..." [Ans: N/A]

3.3 Chemistry In Everyday Life — Core Principles & Governing Laws (Very High Yield — 15 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Chemistry In Everyday Life.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Chemistry In Everyday Life PYQs.

Representative Source PYQs:

- [2024 Morning Shift] Q#1 (MCQ): "Match List I with List II LIST I (Compound) LIST II (Uses) A. Iodoform I. Fire extinguisher B. Carbon tetrachloride II. Insecticide C. CFC II..." [Ans: B]
- [2023 Morning Shift] Q#3 (MCQ): "Which of the following artificial sweeteners has the highest sweetness value in comparison to cane sugar ?..." [Ans: N/A]
- [2023 Morning Shift] Q#3 (MCQ): "Match List I with List II. List I List II Antimicrobials Names (A) Narrow Spectrum Antibiotic (I) Furacin (B) Antiseptic (II) Sulphur dioxid..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2024 (Online) 6th April Morning Shift (Q#1)

Match List I with List II LIST I (Compound) LIST II (Uses) A. Iodoform I. Fire extinguisher B. Carbon tetrachloride II. Insecticide C. CFC III. Antiseptic D. DDT IV. Refrigerants Choose the correct answer from the options given below:

- (A) -II, B-IV, C-I, D-III
- (B) -III, B-I, C-IV, D-II **[Correct]**
- (C) -III, B-II, C-IV, D-I
- (D) -I, B-II, C-III, D-IV

Correct Answer: B

Solution Analysis: The matching for each compound with its respective use is based on their known applications: Iodoform is a compound known for its application as an antiseptic . It has been used historically in medical applications for the treatment of wounds due to its antimicrobial properties. Hence, Iodoform matches with III. Antiseptic . Carbon Tetrachloride was widely used as a fire extinguishing agent due to its property of being a non-flammable liquid that can suffocate fires. However, its use has been significantly reduced in most countries because of its adverse health and environmental impacts. Carbo

Benchmark Problem 2: JEE Main 2024 (Online) 1st February Evening Shift (Q#2)

Match List - I with List - II. List I (Compound) List II (Use) (A) Carbon tetrachloride (I) Paint remover (B) Methylene chloride (II) Refrigerators and air conditioners (C) DDT (III) Fire extinguisher (D) Freons (IV) Non Biodegradable insecticide Choose the correct answer from the options given below :

- (A) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
- (B) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- (C) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
- (D) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Correct Answer: None

Solution Analysis: Each of the compounds listed in List I has distinct uses. Let's match them correctly with the uses mentioned in List II: Carbon tetrachloride (A) was historically used as a cleaning agent, among other things, but due to its toxicity and role in ozone depletion, it is no longer widely used for this purpose; hence it does not match with any of the uses. However, it was previously used in fire extinguishers, so one might think of it in association with (III). Methylene chloride (B) , also known as dichloromethane, is commonly used as a solvent and paint remover, so it matches with (I). DDT (C) is

Benchmark Problem 3: JEE Main 2023 (Online) 8th April Morning Shift (Q#1)

Match List I with List II: LIST I LIST II A. Saccharin I. High potency sweetener B. Aspartame II. First artificial sweetening agent C. Alitame III. Stable at cooking temperature D. Sucralose IV. Unstable at cooking temperature Choose the correct answer from the options given below:

- (A) -II, B-III, C-IV, D-I
- (B) -II, B-IV, C-I, D-III
- (C) -II, B-IV, C-III, D-I
- (D) -IV, B-III, C-I, D-II

Correct Answer: None

Solution Analysis: A. Saccharin is first popular artificial sweetening agent. It is 550 times as sweet as cane sugar. It is used as intake by diabetic person. B. Aspartame is 100 times as sweet as cane sugar. It is used in cold foods and soft drinks because it is unstable at cooking temperature. C. Alitame is 2000 times as sweet compared to cane sugar. It has excellent stability at high temperature. So, can be used in cooking and baking. D. Sucralose is 600 times as sweet as cane sugar. It is heat stable and finds its use in baked goods.